

Foot Traffic Analysis using Mall Dataset and CSRNet Model

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1. Introduction

Retail businesses need to understand how customers move through their stores. Knowing which areas get the most visitors, when traffic peaks, and how people flow through the space helps store managers make better decisions about product placement, staff allocation, store layout, and marketing campaigns.

Foot traffic analysis is the process of automatically measuring and understanding how many people are in a space and where they are standing at any given moment. A computer vision system can analyze CCTV footage in real time or as a pre-recorded video and produce this information automatically.

The key outputs of a foot traffic system are:

- **Person count per frame:** how many people are visible at a specific moment
- **Density maps:** where exactly people are standing in the scene
- **Accumulated heatmaps:** which areas of the store are visited most over time
- **Zone analysis:** breaking the store into sections and measuring traffic per section

This kind of system is already used by major retailers like Walmart, Amazon Go, and Zara to optimize their store operations. Building it from scratch requires solving two separate problems: detecting and localizing people in crowded scenes, and aggregating that information over time into meaningful insights.

1.2. Comparing Approaches

Foot traffic analysis is not a new problem, and several technologies have been used to solve it before computer vision became practical. Each approach has real tradeoffs.

WiFi and Bluetooth Tracking: Retail stores can detect customer smartphones via WiFi or Bluetooth signals. Every phone constantly broadcasts a unique identifier searching for known networks, these signals can be picked up by sensors placed around the store to estimate how many devices (and therefore people) are in each zone.

Limitations:

- Only counts people carrying active smartphones, and maybe someone carries more than one smartphone, therefore he will be counted as 2 persons
- Requires customers to have WiFi or Bluetooth turned on

Infrared and Thermal Sensors: Infrared beam sensors placed at store entrances count how many times the beam is broken given a simple entry/exit count. Thermal cameras detect body heat signatures and can count people in a space.

Limitations:

- Beam sensors only count at entry/exit points, there will be no information about where people go inside the store

- Thermal cameras are expensive and give coarse spatial resolution

1.3. Why Computer Vision

Computer vision using existing CCTV infrastructure offers decisive advantages over the above approaches:

Uses existing hardware: Almost every retail store already has CCTV cameras installed for security purposes. A computer vision system runs on top of this existing infrastructure, no new sensors, no new hardware installation, and no store renovation required.

Rich spatial information: Unlike WiFi counting or beam sensors, computer vision produces a full density map showing exactly where in the store people are standing at every moment. This is what enables heatmap generation and zone analysis.

No customer participation required: WiFi or Bluetooth tracking requires customers to have their signals turned on. Computer vision works regardless of what technology customers carry.

Scalable: The same model can be deployed across multiple stores and multiple cameras with no additional hardware cost only.

Rich output: Beyond just counting people, the density map output enables heatmap generation, zone dwell time analysis, crowd flow direction, and peak hour identification, none of which are possible with sensor-based approaches.

2. Challenges

The core challenge in foot traffic analysis is crowd counting and localization from a fixed overhead camera, this introduces several challenges like:

- People overlap and occlude each other in crowds
- The same camera angle makes people at different distances appear in different scales
- Beside counting the number of people, we need spatial information about their position across a specific interval of time

The first approach that would cross our minds is to use object detection (like YOLO) to draw a bounding box around each person and count the boxes. This works well in sparse scenes but fails in crowds where people overlap heavily, as bounding boxes merge, as a result, detections are missed and the accuracy drops significantly.

The approach I followed instead uses density map estimation, which proved to handle crowds much more gracefully.

3. The Mall Dataset

The Mall Dataset contains 2000 sequential frames extracted from continuous video footage filmed inside a real shopping mall from a ceiling-mounted CCTV camera, exactly matching the deployment scenario for a retail foot traffic system, and that is only one reason I chose this dataset to be the primary training data for.

3.1. What makes it Different from other Datasets

The Two Annotation Files

The Mall Dataset comes with two MATLAB .mat files that are both essential for production-quality training:

mall_gt.mat (Ground Truth Annotations)

This file contains for each of the 2000 frames:

- The (x, y) head position of every person in the frame
- The total person count per frame

Head positions rather than bounding boxes are used because annotating full bounding boxes in crowded scenes is extremely time-consuming and imprecise. A single dot on each person's head is sufficient to know where they are.

perspective_roi.mat (Perspective and Region of Interest)

This file contains two critical maps:

ROI Mask: A binary map (0 or 1) the same dimensions as the frame. Pixels set to 1 represent the walkable floor area where people can actually stand. Pixels set to 0 represent walls, ceilings, shop fronts, and other regions where no person can ever appear. Applying this mask during training tells the model to completely ignore those regions, preventing it from wasting capacity learning that walls have zero people, and preventing false detections from reflections or shadows on non-floor surfaces.

Perspective Map (pMapN): A floating point map the same size as the frame where each pixel value represents the camera's perspective scaling factor at that location. In a ceiling-mounted camera, people standing far away (top of frame) appear much smaller than people standing close to the camera (bottom of frame). The perspective map balances this scaling, as it contains values range from 0.773 (close to camera) to 14.843 (far from camera).

This map is used to make the Gaussian blobs, people far from the camera get a tighter, smaller blob while people close to the camera get a wider blob. Without this correction, the density maps would be spatially misleading and the model would learn a biased representation of crowd density.

4. Density Map Generation

Instead of training the model to output a single count number, I train it to output a density map, a 2D array the same size as the input frame where:

- Every pixel value represents how much "crowd density" exists at that location
- The sum of all pixel values equals the total person count in the frame
- High values (shown as red in a jet colormap) indicate where people are concentrated

By using density maps, the model learns not just how many people there are, but where they are.

4.1. Gaussian Blob

For each annotated head position, I place a 2D Gaussian blob centered at that position. The blob has a peak value of 1.0 at the center and falls off smoothly in all directions. The key parameter is sigma

(σ) which controls how wide the blob spreads, computed adaptively per person using the perspective map, where:

People far from camera $\rightarrow$ high pmap value $\rightarrow$ small sigma $\rightarrow$ tight blob

People close to camera $\rightarrow$ low pmap value $\rightarrow$ large sigma $\rightarrow$ wide blob

After placing all Gaussian blobs and summing them, the entire density map is normalized so its sum equals exactly the number of people in that frame, this ensures the model learns to produce density maps where the sum is always interpretable as a person count.

5. The CSRNet Model

I used CSRNet (Congested Scene Recognition Network), introduced in the 2018 CVPR paper "*CSRNet: Dilated Convolutional Neural Networks for Understanding the Highly Congested Scenes*" by Li et al. CSRNet was specifically designed for crowd counting and density map estimation, making it the most appropriate architecture for this problem.

5.1. Architecture Overview:

Frontend: VGG16 (First 13 layers)

The frontend uses the first 13 layers of VGG16 pretrained on ImageNet, stopping just before the 4th pooling layer. These layers act as a powerful feature extractor, they already understand edges, textures, and object shapes from ImageNet pretraining

Backend: Dilated Convolutions

The backend uses 6 convolutional layers with dilation rate = 2. Dilated convolutions expand the receptive field without reducing spatial resolution.

Output Layer

The final layer produces a single-channel output (the density map prediction) using linear activation. Using relu here was tried initially but caused the model to predict all zeros.

UpSampling2D vs Conv2DTranspose

I experimented with both upsampling approaches:

UpSampling2D: The backend output size is 40×52 ($8 \times$ smaller than input size due to 3 pooling layers). The UpSampling2D(size=(8,8)) layer restores the output to the full 320×416 resolution.

Conv2DTranspose: Unlike the UpSampling2D layer, the Conv2DTranspose layer has trainable parameters that can produce sharper, more accurate upsampled maps.

While the Conv2DTranspose seems better, its performance was worse (MAE 2.97 vs 2.46 with UpSampling2D). This is likely because with only 2000 training frames, the extra parameters in Conv2DTranspose led to overfitting in the upsampling step. UpSampling2D was the better choice given the dataset size.

6. Loss Function and Metrics

Hybrid Loss: MSE + SSIM

While the original CSRNet architecture relies on Mean Squared Error (MSE) for training and Mean Absolute Error (MAE) for evaluation, my implementation introduces a hybrid loss function by combining the Structural Similarity Index Measure (SSIM) with MSE.

$\text{loss} = (1 - \alpha) \times \text{MSE} + \alpha \times (1 - \text{SSIM})$ — $\alpha = 0.84$ (from the MS-SSIM paper)

Relying solely on MSE assumes pixel-wise independence and averages out spatial errors, which often results in overly smooth or blurry density maps that lose distinct crowd features. SSIM measures structural similarity between the predicted and true density maps, it compares luminance, contrast, and spatial patterns rather than just raw pixel values. This hybrid approach ensures that the predicted map not only maintains the correct overall pixel sums (accurate counting via MSE) but also preserves the sharp, distinct peaks of individual heads and structural groupings in dense regions. The weight $\alpha=0.84$ means SSIM contributes 84% of the loss and MSE contributes 16%. This weighting comes from the original MS-SSIM paper and reflects the fact that structural accuracy is harder to learn and needs more gradient pressure than value accuracy.

The following figures illustrate the difference:

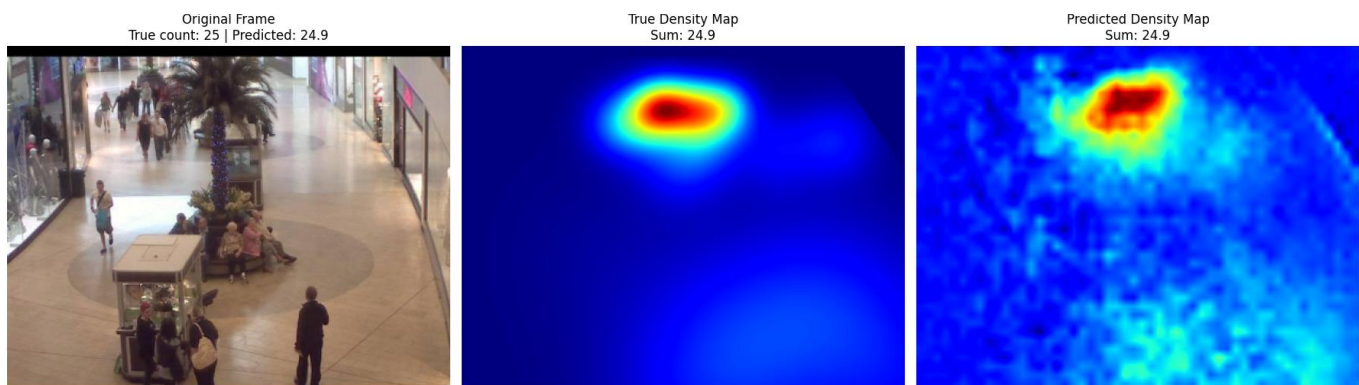


Figure 1: Density map prediction using MSE only

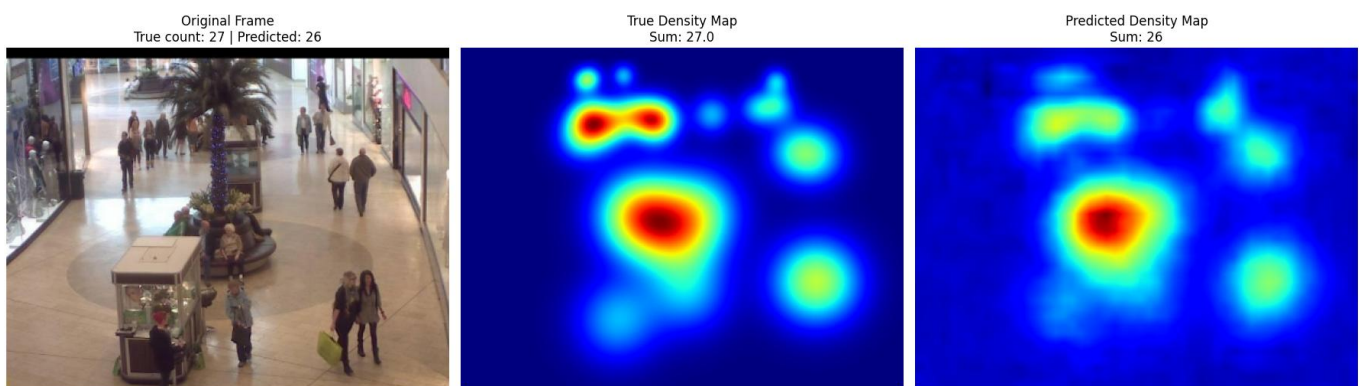


Figure 2: Density map prediction using MSE+SSIM

Evaluation Metric

During training we monitor:

- **Pixel-level MAE:** average absolute error per pixel

For final evaluation we compute:

- **Count MAE:** mean absolute error in person count: $\text{mean}(|\text{predicted_count} - \text{true_count}|)$
- **Count RMSE:** square root mean squared error in person count, more sensitive to large errors

These are computed by summing the predicted density map and dividing by the scale factor to get the predicted person count, then comparing to the ground truth count.

8. Results

Final Performance

Test MAE : 2.46 people per frame

Test RMSE : 3.07 people per frame

Comparison With the Original CSRNet Paper

The original CSRNet paper reports on the Mall Dataset:

Paper MAE : 2.33 people

Paper RMSE : 2.96 people

The results are very close to the paper despite using reduced resolution (320×416 vs 480×640) and only 2000 training frames (CSRNet architecture was trained and evaluated on massive, highly congested benchmark datasets, such as ShanghaiTech and UCF_CC_50). The small gap can be explained by the resolution reduction, at full resolution the results would likely exceed the paper.

9. Potential Improvements

Higher Resolution: Training at full 480×640 would close the remaining gap with the paper's results or exceed it.

Upgrading the Backbone: Replacing the VGG-16 front-end backbone with a more modern architecture (such as ResNet or HRNet) would allow for deeper feature extraction, and maintaining the high-resolution representations that are crucial for detecting small objects like heads in dense crowds.

More Training Data: 2000 frames from a single mall is limited. Adding the MOT17-04 sequence (also filmed in a shopping mall corridor) or ShanghaiTech Part B would expose the model to more diverse scenes and improve generalization.

Data Augmentation: To further improve the model's robustness for deployment, the data augmentation pipeline incorporates random variations in brightness and contrast. In a real-world scenario, lighting conditions are rarely static, The environment undergoes illumination shifts throughout the day.

Temporal Information: CSRNet processes each frame independently. A recurrent or 3D convolutional model could use information from previous frames to improve predictions, if someone was in a location one frame ago, they are likely still nearby.

Adaptive Perspective Map: Current solution mitigates perspective distortion using a static perspective map (perspective_roi.mat) to scale the regions of interest. Alternative, more dynamic solutions could be implemented to handle this issue natively within the architecture.